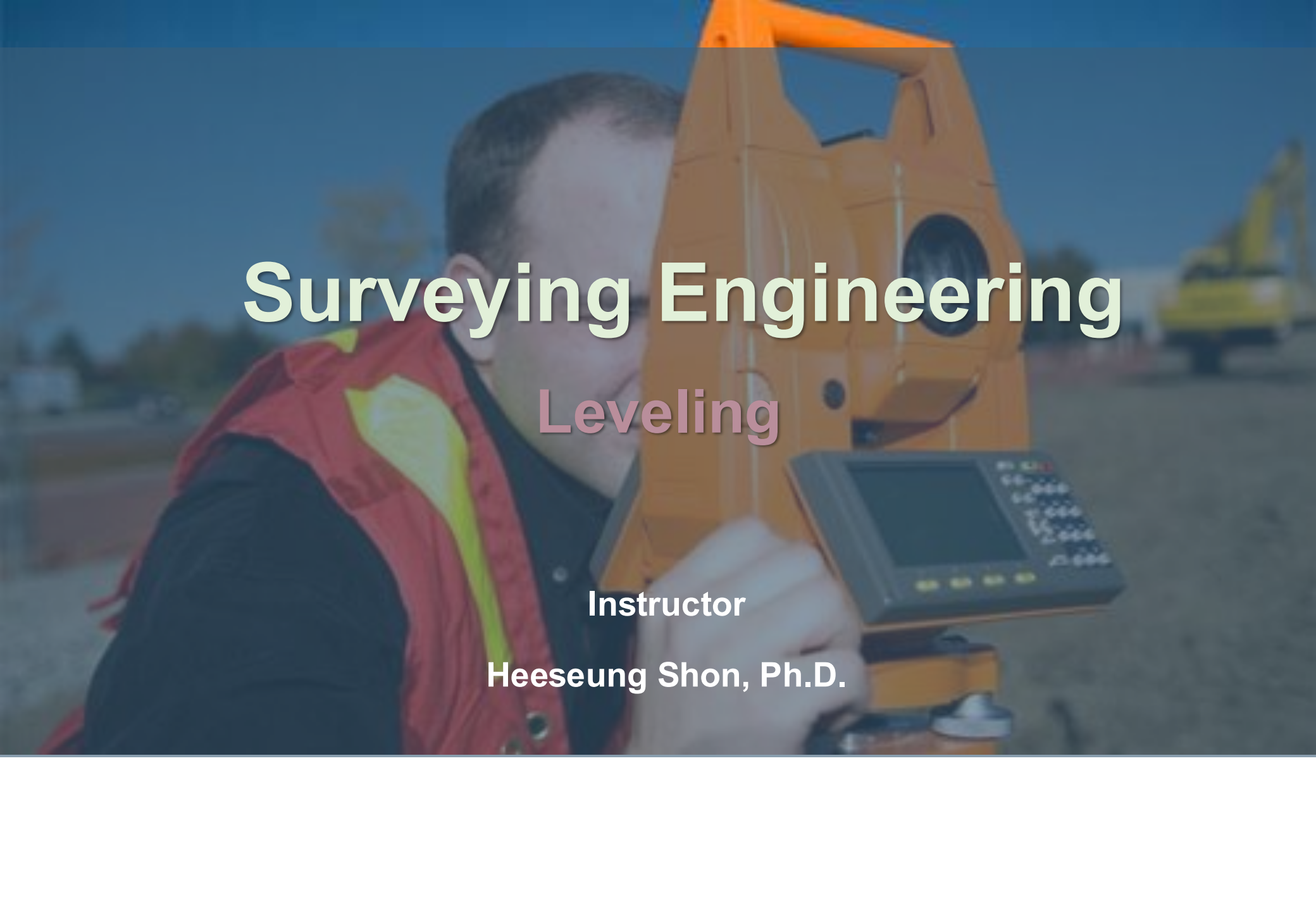


A surveyor wearing a red safety vest is operating a yellow total station instrument on a construction site. The background shows a blurred yellow excavator and a clear blue sky.

Surveying Engineering

Leveling

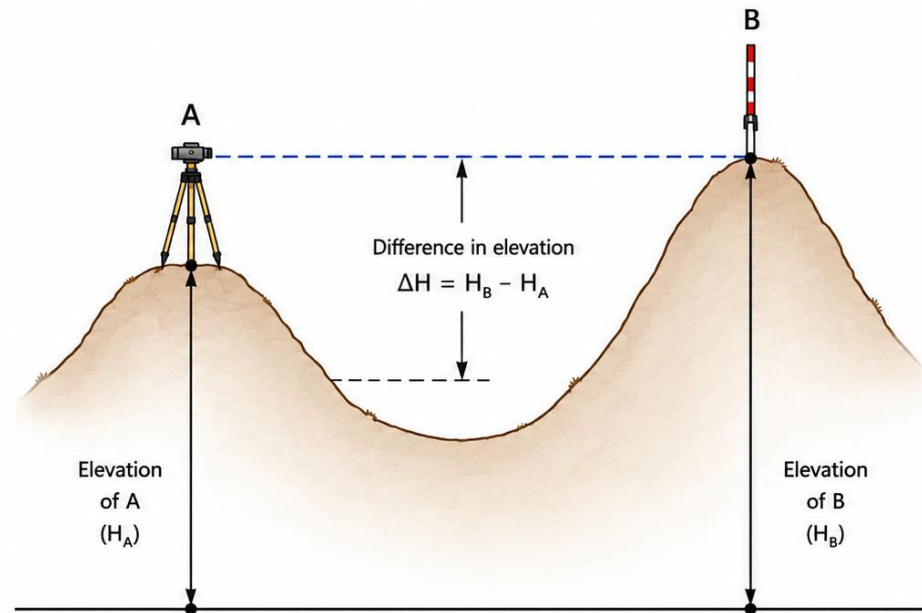
Instructor

Heeseung Shon, Ph.D.

Leveling and Vertical Reference Surfaces

□ Definition of Leveling

- The elevation of a point is its vertical position relative to a chosen reference level surface.
- Leveling determines the elevations of points, or the difference in elevation between points, by direct or indirect measurement.

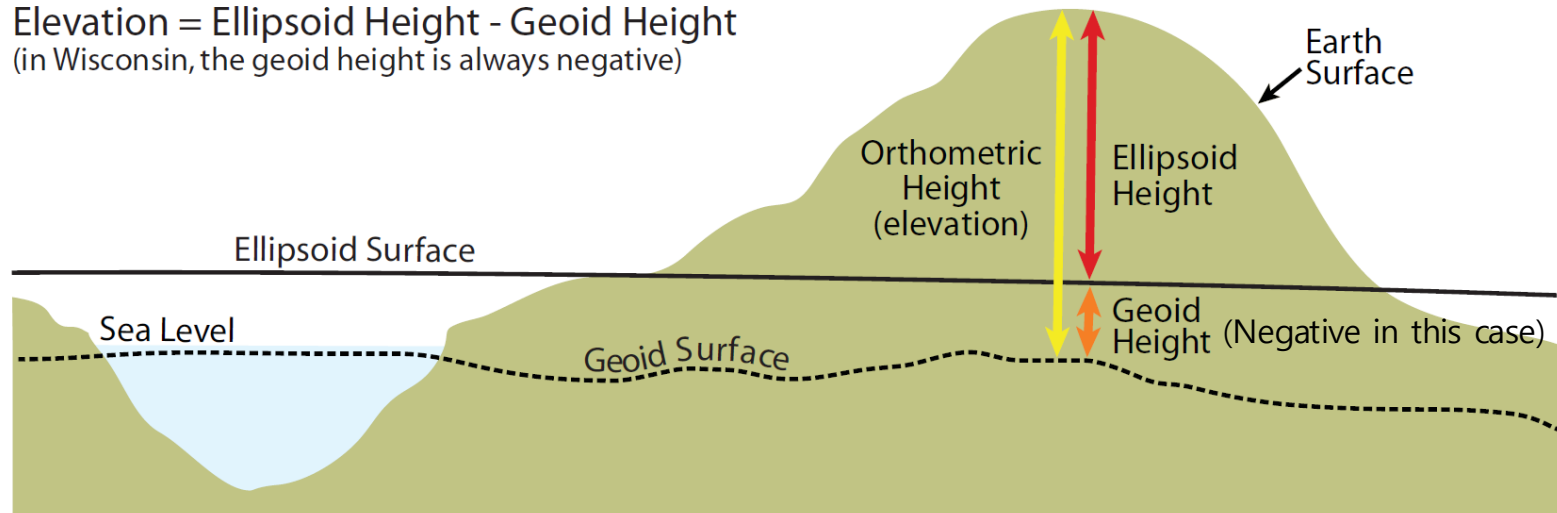


Leveling and Vertical Reference Surfaces

□ Vertical Surfaces and Datums

- **Topography:** the actual, irregular physical surface of the Earth, including natural and man-made features.
- **The Geoid:** the level surface that coincides with mean sea level extended continently.
- **The Ellipsoid:** a smooth mathematical surface that approximates the shape of the Earth and is used a reference for computations and positioning.

Elevation = Ellipsoid Height - Geoid Height
(in Wisconsin, the geoid height is always negative)

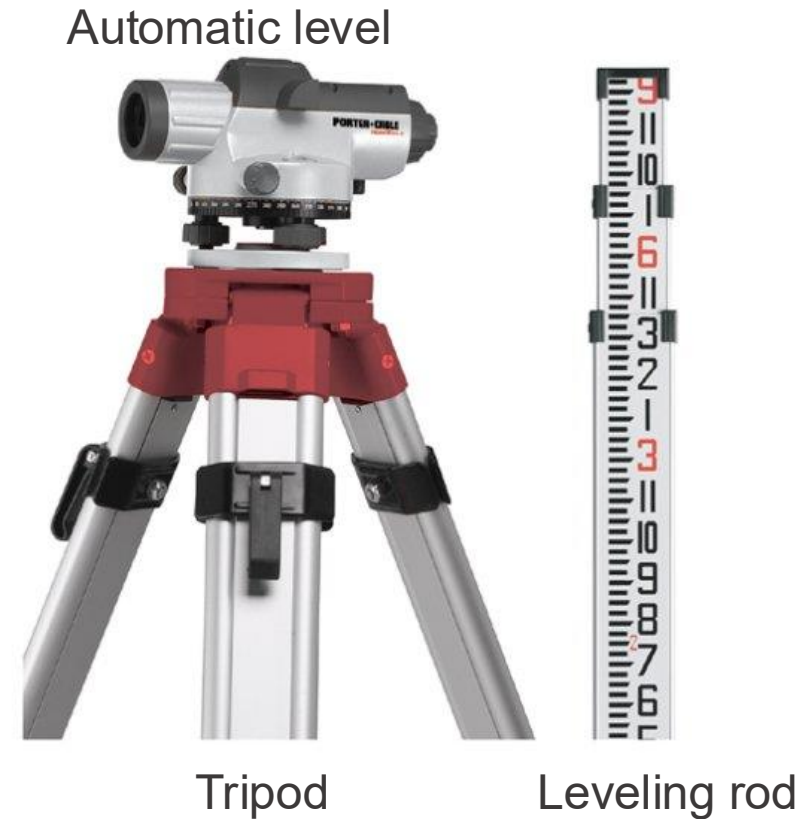


Automatic Level

- An automatic level uses an internal compensator to establish a horizontal line of sight.
- After the bull's-eye bubble is approximately centered, the compensator automatically corrects small instrument tilts.
- Automatic levels are widely used for general leveling and elevation measurements.

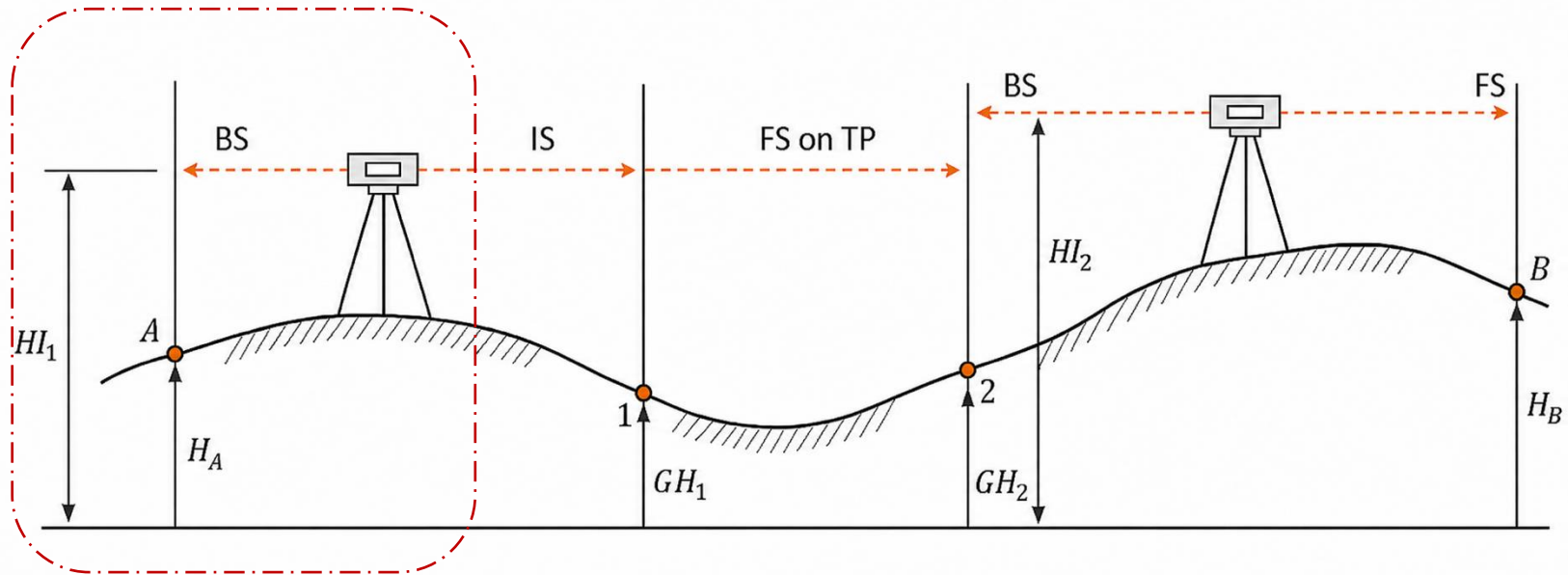


Bull's-eye bubble



Automatic Level

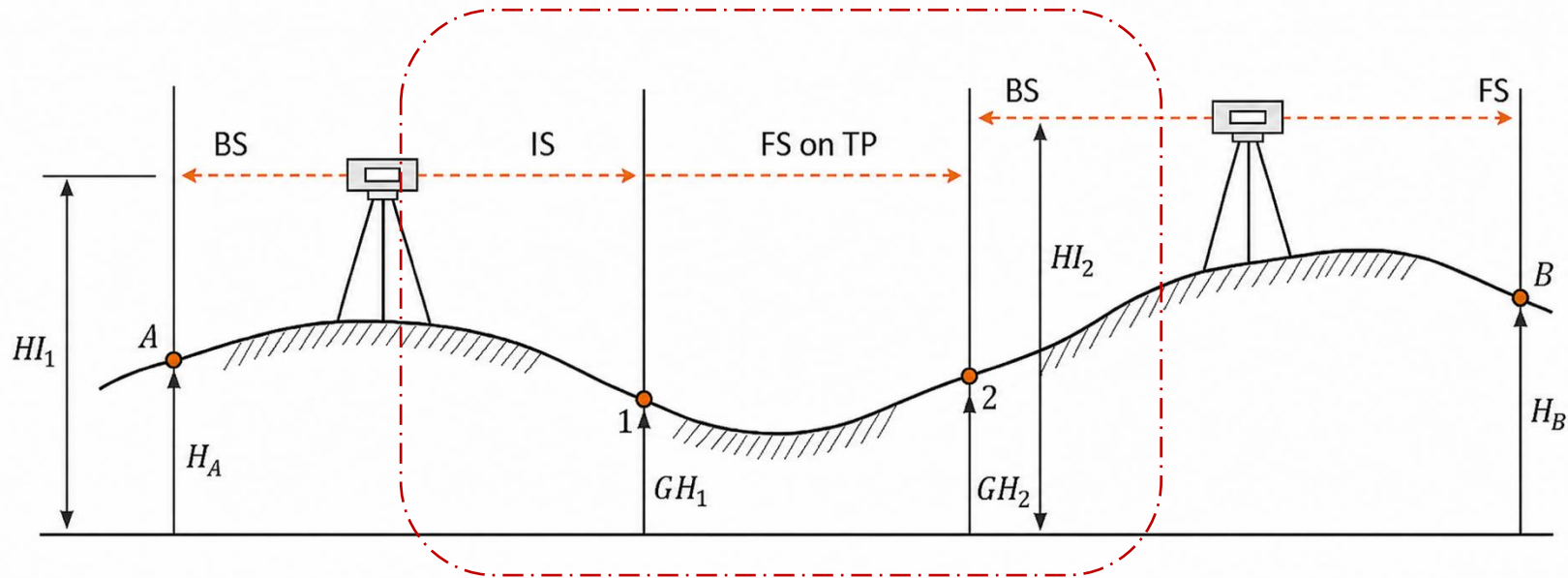
Terminology



- **A**: The starting point of the leveling survey. Its elevation is known.
- H_A : The elevation of Point A above the selected reference datum.
- **BS**: Backsight - the first rod reading taken after setting up the instrument.
- $HI_1 = H_A + BS$: the elevation of the horizontal line of sight

Automatic Level

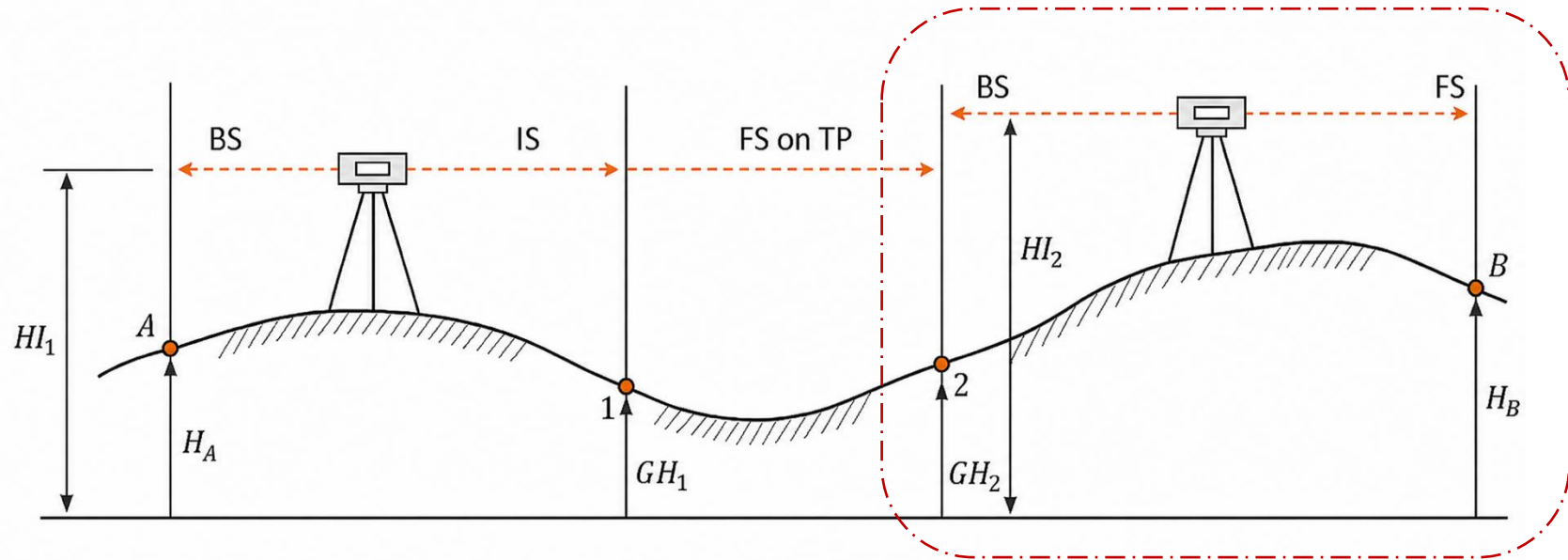
Terminology



- **IS**: intermediate sight - a rod reading taken on a point whose elevation is required.
- $GH_1 = HI_1 - IS$: ground elevation
- **FS on TP** : Foresight on turning point - a stable point used to transfer elevation between two instrument setups.
- $GH_2 = HI_1 - FS \text{ on TP}$: ground elevation

Automatic Level

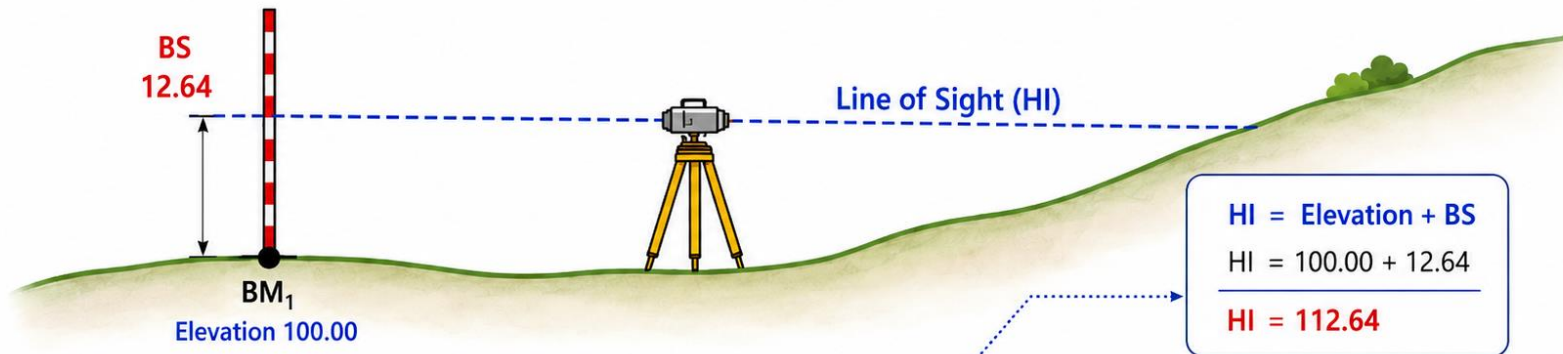
Terminology



- **BS**: Backsight - the first rod reading taken after setting up the instrument.
- $HI_2 = GH_2 + BS$: the elevation of the horizontal line of sight
- **FS**: Foresight - the last rod reading taken before moving the instrument.
- $H_B = HI_2 - FS$: elevation of the target point B

Automatic Level

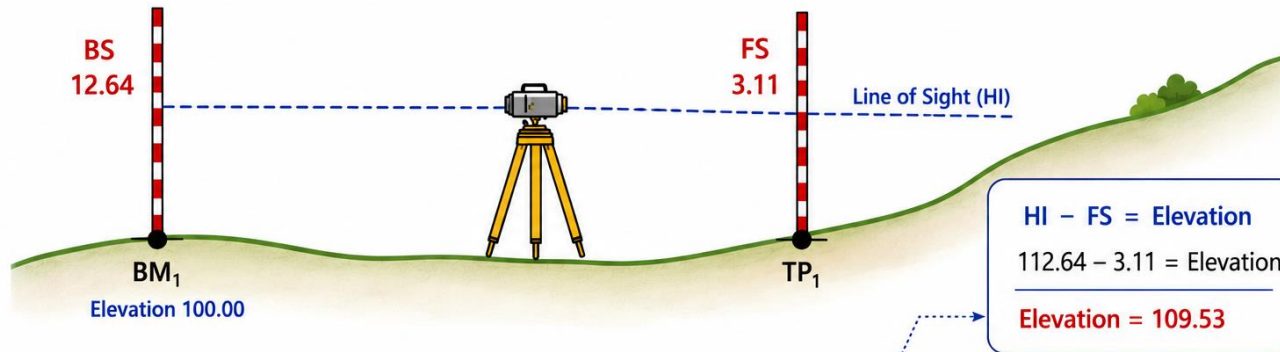
□ Example



Point	BS	HI	FS	Elevation
BM ₁	12.64	112.64		100.00

Automatic Level

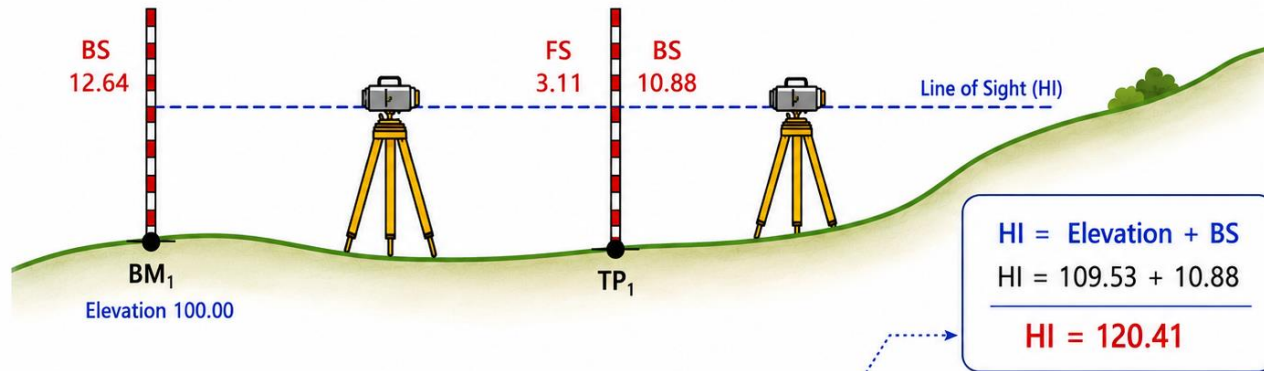
□ Example



Point	BS	HI	FS	Elevation
BM ₁	12.64	112.64		100.00
TP ₁			3.11	109.53

Automatic Level

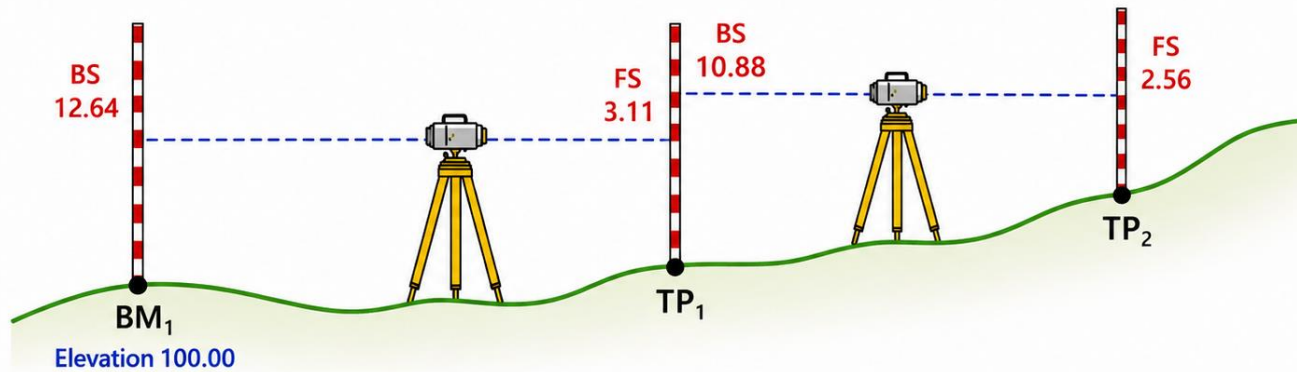
□ Example



Point	BS	HI	FS	Elevation
BM ₁	12.64	112.64		100.00
TP ₁	10.88	120.41	3.11	109.53

Automatic Level

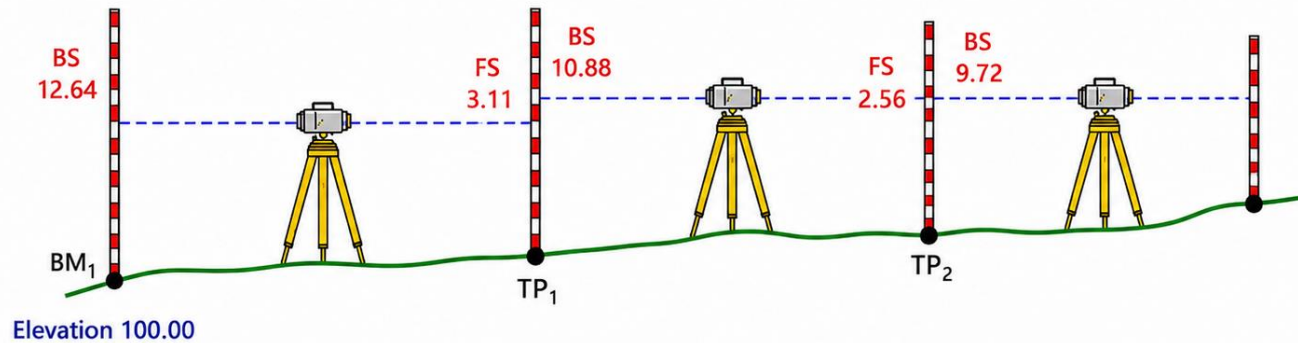
□ Example



Point	BS	HI	FS	Elevation
BM ₁	12.64	112.64		100.00
TP ₁	10.88	120.41	3.11	109.53
TP ₂			2.56	117.85

Automatic Level

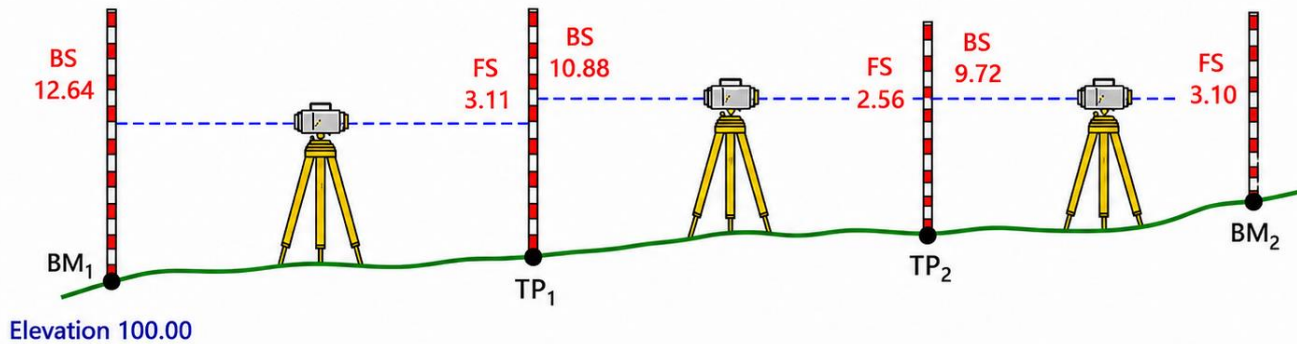
□ Example



Point	BS	HI	FS	Elevation
BM ₁	12.64	112.64		100.00
TP ₁	10.88	120.41	3.11	109.53
TP ₂	9.72	127.57	2.56	117.85

Automatic Level

□ Example



Point	BS	HI	FS	Elevation
BM ₁	12.64	112.64		100.00
TP ₁	10.88	120.41	3.11	109.53
TP ₂	9.72	127.57	2.56	117.85
BM ₂			3.10	124.47

Precision

❑ Source of mistakes

- Misreading or transposing rod readings
- Recording BS values in the FS column, or vice versa
- Arithmetic errors
- Using different rod positions for FS and BS at a turning point
- Reading a telescopic rod before it is fully extended

Precision

❑ Source of errors

- Instrument maladjustment
- Parallax caused by improper focusing
- Earth curvature
- Atmospheric refraction
- Temperature-induced changes in the instrument or rod
- Expansion or contraction of the leveling rod
- Unstable turning points
- Bubble not centered at the time of sighting